

HINTS AND SOLUTIONS

1. (b) We know that, π radian = 180°
 $\Rightarrow 1 \text{ radian} = \frac{180^\circ}{\pi} = \frac{180^\circ}{22} \times 7$
 $= \frac{630^\circ}{11} = 57\frac{3^\circ}{11} = 57^\circ + \frac{3 \times 60}{11} \text{ min}$
 $= 57^\circ + 16' + \frac{4}{11} \text{ min}$
 $= 57^\circ + 16' + \frac{4}{11} \times 60 \text{ s}$
 $= 57^\circ + 16' + 21.8''$
 $= 57^\circ 16' 21.8'' = 57^\circ 16' 22''$

2. (a) $-47^\circ 30' = -\left(47 + \frac{30}{60}\right)^\circ$
 $= -\left(47 + \frac{1}{2}\right)^\circ \quad \left[\because 1' = \left(\frac{1}{60}\right)^\circ\right]$
 $= -\left(\frac{94+1}{2}\right)^\circ = \frac{-95^\circ}{2}$
 $= \frac{-95}{2} \times \frac{\pi}{180} = \frac{-19\pi}{72} \text{ rad}$

3. (b) Here, $r = 60 \text{ cm}$, $l = 16.5 \text{ cm}$
 $\therefore \theta = \frac{l}{r} = \left(\frac{16.5}{60}\right)^\circ$
 $= \left(\frac{16.5}{60} \times \frac{180}{\pi}\right)^\circ = \left(\frac{16.5}{60} \times \frac{180}{22} \times 7\right)^\circ$
 $= \left(\frac{63}{4}\right)^\circ = 15^\circ 45'$

4. (c) Speed of train = 90 km/h
 $= \left(90 \times \frac{5}{18}\right) \text{ m/sec} = 25 \text{ m/sec}$
 Distance moved in 11 sec
 $= (25 \times 11) \text{ m} = 275 \text{ m}$
 $\therefore l = 275 \text{ m}$, $r = 1500 \text{ m}$
 $\therefore \theta = \frac{l}{r} \Rightarrow \theta = \left(\frac{275}{1500}\right)^\circ$
 $= \left(\frac{275}{1500} \times \frac{180}{\pi}\right)^\circ = \left(\frac{275}{1500} \times \frac{180}{22} \times 7\right)^\circ$
 $= \left(\frac{21}{2}\right)^\circ = 10^\circ 30'$

5. (b) Angle traced by hour hand in 1 hr = 30°
 Angle traced by hour hand in $\frac{91}{12}$ hrs = $\left(30 \times \frac{91}{12}\right)^\circ = 227^\circ 30'$
 Angle traced by minute hand in 1 min = 6°
 Angle traced by minute hand in 35 min = $(6 \times 35)^\circ = 210^\circ$
 $\therefore$ Required angle = $227^\circ 30' - 210^\circ$
 $= 17^\circ 30'$

6. (b) Let the hands of a clock coincide after ' t ' min.

Angle traced by hour hand in

$$t \text{ min} = \frac{1}{2}t$$

Angle traced by minute hand in

$$t \text{ min} = 6t$$

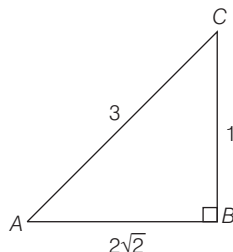
Angle between hour hand and minute hand at 5 pm = $5 \times 30 = 150^\circ$

$$\therefore 6t - \frac{1}{2}t = 150^\circ \Rightarrow t = \frac{150 \times 2}{11} = \frac{300}{11}$$

$$= 27 + \frac{3}{11} = 27 \text{ min } 16 \text{ sec}$$

Therefore, required time = 5:27:16 pm

7. (a) Given, $\sin A = \frac{1}{3}$



In right angled $\triangle ABC$,

$$AB^2 = AC^2 - BC^2 = 3^2 - 1^2 = 8$$

$$\Rightarrow AB = 2\sqrt{2}$$

$$\therefore \cos A = \frac{AB}{AC} = \frac{2\sqrt{2}}{3}$$

$$\text{and } \sec A = \frac{AC}{AB} = \frac{3}{2\sqrt{2}}$$

$$\tan A = \frac{BC}{AB} = \frac{1}{2\sqrt{2}}$$

$$\text{and } \operatorname{cosec} A = \frac{AC}{BC} = 3$$

$$\therefore \cos A \cdot \operatorname{cosec} A + \tan A \cdot \sec A$$

$$= \frac{2\sqrt{2}}{3} \cdot \frac{3}{1} + \frac{1}{2\sqrt{2}} \cdot \frac{3}{2\sqrt{2}}$$

$$= 2\sqrt{2} + \frac{3}{8} = \frac{16\sqrt{2} + 3}{8}$$

8. (b) $\cos 15^\circ - \sin 15^\circ = \cos 15^\circ - \sin (90^\circ - 75^\circ)$

$$= \cos 15^\circ - \cos 75^\circ$$

$$= 2 \sin \frac{15^\circ + 75^\circ}{2} \cdot \sin \frac{75^\circ - 15^\circ}{2}$$

$$\left(\because \cos C - \cos D = 2 \sin \frac{C+D}{2} \sin \frac{D-C}{2} \right)$$

$$= 2 \sin 45^\circ \cdot \sin 30^\circ = 2 \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{1}{\sqrt{2}}$$

9. (c) Given, $3 \tan \theta = 4 \Rightarrow \tan \theta = \frac{4}{3}$

$$\sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} \times \frac{1-\sin \theta}{1-\sin \theta}$$

$$= \frac{1-\sin \theta}{\sqrt{1-\sin^2 \theta}}$$

$$= \frac{1-\sin \theta}{\cos \theta} = \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \sec \theta - \tan \theta$$

$$= \sqrt{1+\tan^2 \theta} - \tan \theta$$

$$= \sqrt{1+\left(\frac{4}{3}\right)^2} - \frac{4}{3}$$

$$= \sqrt{1+\frac{16}{9}} - \frac{4}{3} = \frac{5}{3} - \frac{4}{3} = \frac{1}{3}$$

10. (b) Given, $\tan A = 1 = \tan 45^\circ$

$$\Rightarrow A = 45^\circ \text{ and } \tan B = \sqrt{3} = \tan 60^\circ$$

$$\therefore B = 60^\circ$$

Now, $\cos A \cos B - \sin A \sin B$

$$= \cos 45^\circ \cos 60^\circ - \sin 45^\circ \sin 60^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{2} - \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{1-\sqrt{3}}{2\sqrt{2}}$$

11. (b) $\operatorname{cosec}^2 \theta - 2 + \sin^2 \theta$

$$= (\sin \theta - \operatorname{cosec} \theta)^2$$

Hence, it is always non-negative.

12. (b) $\sin^2 15^\circ + \sin^2 20^\circ + \sin^2 25^\circ$

$$+ \dots + \sin^2 75^\circ$$

$$= (\sin^2 15^\circ + \sin^2 75^\circ)$$

$$+ (\sin^2 20^\circ + \sin^2 70^\circ) + \dots + \sin^2 45^\circ$$

$$= [\sin^2 15^\circ + \sin^2 (90^\circ - 15^\circ)]$$

$$+ [\sin^2 20^\circ + \sin^2 (90^\circ - 20^\circ)] + \dots + \sin^2 45^\circ$$

$$= (\sin^2 15^\circ + \cos^2 15^\circ)$$

$$+ (\sin^2 20^\circ + \cos^2 20^\circ) + \dots + 1/2$$

$$= 6 + 1/2 = 13/2$$

$$(\because \sin^2 \theta + \cos^2 \theta = 1)$$

13. (b) $\frac{5 \sin 75^\circ \sin 77^\circ + 2 \cos 13^\circ \cos 15^\circ}{\cos 15^\circ \sin 77^\circ}$

$$- \frac{7 \sin 81^\circ}{\cos 9^\circ}$$

$$= \frac{5 \cos 15^\circ \sin 77^\circ + 2 \sin 77^\circ \cos 15^\circ}{\cos 15^\circ \sin 77^\circ}$$

$$- \frac{7 \cos 9^\circ}{\cos 9^\circ}$$

$$= \frac{7 \cos 15^\circ \cdot \sin 77^\circ}{\cos 15^\circ \cdot \sin 77^\circ} - \frac{7 \cos 9^\circ}{\cos 9^\circ}$$

$$= 7 - 7 = 0$$

$$\begin{aligned}
 14. (a) & \log(\tan 1^\circ) + \log(\tan 2^\circ) + \dots + \log(\tan 89^\circ) \\
 &= \log(\tan 1^\circ \tan 2^\circ \dots \tan 45^\circ \dots \tan 88^\circ \tan 89^\circ) \\
 &= \log[(\tan 1^\circ \cot 1^\circ)(\tan 2^\circ \cot 2^\circ) \dots \tan 45^\circ] \\
 &= \log(1^\circ \cdot 1^\circ \dots 1^\circ) = 0 \quad [\because \tan 89^\circ = \tan(90^\circ - 1^\circ) = \cot 1^\circ]
 \end{aligned}$$

$$\begin{aligned}
 15. (c) & \because \text{Given, } \tan \theta + \frac{1}{\tan \theta} = 2 \\
 & \text{On squaring both side, we get } \left(\tan \theta + \frac{1}{\tan \theta} \right)^2 = 2^2 \\
 & \Rightarrow \tan^2 \theta + \frac{1}{\tan^2 \theta} + 2 = 4 \Rightarrow \tan^2 \theta + \frac{1}{\tan^2 \theta} = 2
 \end{aligned}$$

$$\begin{aligned}
 16. (c) & \text{Given, } \sec \theta = \frac{13}{5} \\
 \therefore \tan \theta &= \sqrt{\sec^2 \theta - 1} = \sqrt{\frac{169}{25} - 1} = \sqrt{\frac{144}{25}} = \frac{12}{5} \\
 \therefore \frac{2 \sin \theta - 3 \cos \theta}{4 \sin \theta - 9 \cos \theta} &= \frac{\frac{2 \sin \theta}{\cos \theta} - 3}{4 \frac{\sin \theta}{\cos \theta} - 9} \\
 & \quad \text{(dividing numerator and denominator by } \cos \theta) \\
 &= \frac{2 \tan \theta - 3}{4 \tan \theta - 9} = \frac{2 \times \frac{12}{5} - 3}{4 \times \frac{12}{5} - 9} = \frac{9/5}{3/5} = \frac{9}{3} = 3
 \end{aligned}$$

$$\begin{aligned}
 17. (c) & \sin(45^\circ + A) - \cos(45^\circ - A) \\
 &= \sin(45^\circ + A) - \cos[90^\circ - (45^\circ + A)] = \sin(45^\circ + A) - \sin(45^\circ + A) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 18. (b) & \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}} = \sqrt{\frac{(1 + \sin \theta)(1 + \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)}} \\
 &= \sqrt{\frac{(1 + \sin \theta)^2}{1 - \sin^2 \theta}} = \sqrt{\frac{(1 + \sin \theta)^2}{\cos^2 \theta}} = \frac{1 + \sin \theta}{\cos \theta} = \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} \\
 &= \sec \theta + \tan \theta
 \end{aligned}$$

$$\begin{aligned}
 19. (b) & \sin(n+1)A \sin(n-1)A + \cos(n+1)A \cos(n-1)A \\
 &= \cos[(n+1)A - (n-1)A] \\
 & \quad [\because \cos A \cos B + \sin A \sin B = \cos(A - B)] \\
 &= \cos[nA + A - nA + A] = \cos 2A
 \end{aligned}$$

$$\begin{aligned}
 20. (a) & \frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ - \sin 11^\circ} \\
 & \text{Divide numerator and denominator by } \cos 11^\circ \\
 &= \frac{1 + \tan 11^\circ}{1 - \tan 11^\circ} = \frac{\tan 45^\circ + \tan 11^\circ}{1 - \tan 45^\circ \tan 11^\circ} = \tan(45^\circ + 11^\circ) = \tan 56^\circ
 \end{aligned}$$

$$\begin{aligned}
 21. (c) & \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}} = \frac{\frac{a^2 - b^2}{a^2 + b^2}}{\sqrt{1 - \frac{(a^2 - b^2)^2}{(a^2 + b^2)^2}}} \\
 &= \frac{\frac{a^2 - b^2}{a^2 + b^2}}{\frac{\sqrt{(a^2 + b^2)^2 - (a^2 - b^2)^2}}{a^2 + b^2}} = \frac{a^2 - b^2}{\sqrt{4a^2 b^2}} = \frac{a^2 - b^2}{2ab}
 \end{aligned}$$

$$\begin{aligned}
 22. (d) & \text{Given, } \cos \theta = 0.96, \sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - (0.96)^2} \\
 &= \sqrt{1 - 0.9216} = 0.28, \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{0.28}{0.96} = \frac{7}{24} \\
 & \frac{1}{\sin \theta} + \frac{1}{\tan \theta} = \frac{100}{28} + \frac{24}{7} = \frac{196}{28} = 7
 \end{aligned}$$

$$23. (b) \text{ As we know, } \sin x \text{ is increasing from } 0 \text{ to } 90^\circ.$$

$$\therefore \sin y > \sin x$$

$$24. (a) \text{ Given, } (\sin \theta + \operatorname{cosec} \theta) = 2.5$$

$$\begin{aligned}
 \Rightarrow & \left(\sin \theta + \frac{1}{\sin \theta} \right) = \frac{5}{2} \Rightarrow 2 \sin^2 \theta - 5 \sin \theta + 2 = 0 \\
 \Rightarrow & 2 \sin^2 \theta - 4 \sin \theta - \sin \theta + 2 = 0 \\
 \Rightarrow & 2 \sin \theta (\sin \theta - 2) - 1(\sin \theta - 2) = 0 \\
 \Rightarrow & (2 \sin \theta - 1)(\sin \theta - 2) = 0 \Rightarrow \sin \theta = \frac{1}{2} \quad (\because \sin \theta \neq 2)
 \end{aligned}$$

$$\therefore \theta = 30^\circ$$

$$25. (a) \text{ Given, } \frac{\cos x}{\cos y} = n \text{ and } \frac{\sin x}{\sin y} = m \quad \dots(i)$$

$$\begin{aligned}
 \text{Now, } (m^2 - n^2) \sin^2 y &= \left(\frac{\sin^2 x}{\sin^2 y} - \frac{\cos^2 x}{\cos^2 y} \right) \sin^2 y \\
 &= \frac{(1 - \cos^2 x) \cos^2 y - \cos^2 x (1 - \cos^2 y)}{\cos^2 y} \\
 &= \frac{\cos^2 y - \cos^2 x}{\cos^2 y} = 1 - n^2 \quad [\text{from Eq. (i)}]
 \end{aligned}$$

$$26. (b) \text{ Given, } p = \tan^2 x + \cot^2 x = \tan^2 x + \frac{1}{\tan^2 x}$$

$$\therefore \text{AM} \geq \text{GM} \Rightarrow \tan^2 x + \frac{1}{\tan^2 x} \geq 2 \left(\tan^2 x \cdot \frac{1}{\tan^2 x} \right)^{1/2}$$

$$\Rightarrow \tan^2 x + \frac{1}{\tan^2 x} \geq 2 \Rightarrow p \geq 2$$

$$27. (a) \text{ Let angles in circular measure are } A \text{ and } B, \text{ then degree measures will be } A \left(\frac{180}{\pi} \right) \text{ and } B \left(\frac{180}{\pi} \right).$$

$$\text{By given condition, } A + B = 1 \quad \dots(i)$$

$$A \left(\frac{180}{\pi} \right) - B \left(\frac{180}{\pi} \right) = 1 \quad \dots(ii)$$

$$\text{On multiplying Eq. (i) by } \frac{180}{\pi} \text{ and adding to Eq. (ii), we get}$$

$$2A \left(\frac{180}{\pi} \right) = \frac{180}{\pi} + 1 \Rightarrow A = \frac{1}{2} \times \frac{11}{180} \left[\frac{180}{\pi} + 1 \right] = \frac{1}{2} + \frac{\pi}{360}$$

$$\therefore B = 1 - A = 1 - \left(\frac{1}{2} + \frac{\pi}{360} \right) = \frac{1}{2} - \frac{\pi}{360}$$

$$\begin{aligned}
 28. (b) & [(1 - \sin^2 \theta) \sec^2 \theta + \tan^2 \theta] (\cos^2 \theta + 1) \\
 &= [\cos^2 \theta \cdot \sec^2 \theta + \tan^2 \theta] (\cos^2 \theta + 1) \quad (\because \sin^2 \theta + \cos^2 \theta = 1) \\
 &= (1 + \tan^2 \theta) (\cos^2 \theta + 1) = \sec^2 \theta (\cos^2 \theta + 1) \\
 &= \sec^2 \theta \cdot \cos^2 \theta + \sec^2 \theta = 1 + \sec^2 \theta > 2 \quad (\because \sec^2 \theta - \tan^2 \theta = 1) \\
 & \quad (\because \sec^2 \theta > 1 \text{ for } 0 < \theta < 90^\circ)
 \end{aligned}$$

$$\begin{aligned}
 29. (d) & \text{Given, } 7 \cos^2 \theta + 3 \sin^2 \theta = 4 \Rightarrow 7 \cos^2 \theta + 3(1 - \cos^2 \theta) = 4 \\
 \Rightarrow & 3 - 4 \cos^2 \theta = 4 \Rightarrow \cos^2 \theta = 1/4
 \end{aligned}$$

$$\text{Now, } \tan^2 \theta = \sec^2 \theta - 1 = \frac{1}{\cos^2 \theta} - 1 \Rightarrow \tan^2 \theta = 4 - 1 = 3$$

$$\therefore \tan \theta = \sqrt{3} \quad (\because 0 < \theta < \pi/2)$$

$$30. (d) \text{ We know that, the value of } \cos \theta \text{ is decreasing in the interval } 0 \leq \theta \leq 90^\circ$$

$$\therefore \cos 1^\circ > \cos 89^\circ \Rightarrow p > q$$

Also, $\cos 1^\circ$ is close to 1 and $\cos 89^\circ$ is close to 0.

Hence, option (d) is correct.

31. (d) We know in the interval $\theta \in (0, \pi/2)$, $\sec^2 \theta$ is increasing from 1 to ∞ .

$$\therefore p \geq 1$$

32. (b) $\because \cos 90^\circ = 0$

$$\therefore \cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 90^\circ = 0 \quad (\because \cos 90^\circ = 0)$$

33. (c) Given, $\tan A = \frac{1 - \cos B}{\sin B}$

$$\begin{aligned} \therefore \frac{2 \tan A}{1 - \tan^2 A} &= \frac{2 \cdot \frac{(1 - \cos B)}{\sin B}}{1 - \left(\frac{(1 - \cos B)}{\sin B} \right)^2} \\ &= \frac{2(1 - \cos B) \sin B}{\sin^2 B - 1 - \cos^2 B + 2 \cos B} = \frac{2(1 - \cos B) \sin B}{-2 \cos^2 B + 2 \cos B} \\ &= \frac{2 \sin B (1 - \cos B)}{2 \cos B (1 - \cos B)} = \frac{\sin B}{\cos B} = \tan B \end{aligned}$$

34. (c) $\cot 15^\circ \cot 20^\circ \cot 70^\circ \cot 75^\circ$

$$\begin{aligned} &= \tan(90^\circ - 15^\circ) \tan(90^\circ - 20^\circ) \cot 70^\circ \cot 75^\circ \\ &= \tan 75^\circ \tan 70^\circ \frac{1}{\tan 70^\circ} \cdot \frac{1}{\tan 75^\circ} = 1 \end{aligned}$$

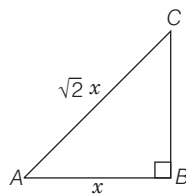
35. (b) Given, $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = 2$

$$\therefore \sin^2 \theta + \cos^2 \theta = 2 \sin \theta \cos \theta$$

$$\Rightarrow \sin 2\theta = 1 = \sin 90^\circ \Rightarrow 2\theta = 90^\circ \Rightarrow \theta = 45^\circ$$

36. (d) Given, in right angled $\triangle ABC$, $\angle B = 90^\circ$ and $AB : AC = 1 : \sqrt{2}$

$$\text{Let } AB = x, AC = \sqrt{2}x$$



$$\Rightarrow AB^2 + BC^2 = AC^2 \quad (\text{using pythagoras theorem})$$

$$\Rightarrow x^2 + BC^2 = (\sqrt{2}x)^2$$

$$\Rightarrow BC^2 = 2x^2 - x^2 \Rightarrow BC = x$$

$$\therefore \tan A = \frac{BC}{AB} = \frac{x}{x} = 1$$

$$\therefore \frac{2 \tan A}{1 - \tan^2 A} = \frac{2(1)}{1 - 1^2} = \frac{2}{0} = \infty \quad (\text{undefined})$$

37. (a) Given, $AD = DB = \frac{a}{2}$ and $\angle B = 90^\circ$

$$\text{In right angled } \triangle ABC, BC^2 + AB^2 = AC^2$$

$$\Rightarrow BC^2 + a^2 = b^2 \Rightarrow BC = \sqrt{b^2 - a^2}$$

$$\text{In right angled } \triangle BCD, CD^2 = BC^2 + BD^2$$

$$CD^2 = (\sqrt{b^2 - a^2})^2 + \left(\frac{a}{2}\right)^2 = \left(b^2 - a^2 + \frac{a^2}{4}\right)$$

$$\Rightarrow CD^2 = \frac{4b^2 - 3a^2}{4} \Rightarrow CD = \frac{\sqrt{4b^2 - 3a^2}}{2}$$

$$\therefore \cos \theta = \frac{BC}{CD} = \frac{\sqrt{b^2 - a^2}}{\frac{\sqrt{4b^2 - 3a^2}}{2}} = \frac{2\sqrt{b^2 - a^2}}{\sqrt{4b^2 - 3a^2}}$$

38. (d) Given, $\sin x + \sqrt{3} \cos x = 1$

$$\text{On dividing both sides by 2, we get } \frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x = \frac{1}{2}$$

$$\Rightarrow \sin x \sin 30^\circ + \cos x \cos 30^\circ = 1/2$$

$$\Rightarrow \cos(x - 30^\circ) = \cos 60^\circ \Rightarrow x - 30^\circ = 60^\circ$$

$$\therefore x = 90^\circ$$

39. (c) $\operatorname{cosec} A \cdot \sec A = \frac{\sqrt{1 + \tan^2 A}}{\tan A} \cdot \sqrt{1 + \tan^2 A}$

$$= \frac{1 + \tan^2 A}{\tan A} = \frac{1 + (\sqrt{2} - 1)^2}{\sqrt{2} - 1} \quad (\because \tan A = \sqrt{2} - 1)$$

$$= \frac{1 + 2 + 1 - 2\sqrt{2}}{\sqrt{2} - 1} = \frac{4 - 2\sqrt{2}}{(\sqrt{2} - 1)} = \frac{2\sqrt{2}(\sqrt{2} - 1)}{(\sqrt{2} - 1)} = 2\sqrt{2}$$

40. (c) Given, $\tan \theta = \frac{3}{1} \Rightarrow \sin \theta = 3 \cos \theta \Rightarrow \sin \theta = 3\sqrt{1 - \sin^2 \theta}$

$$\text{Squaring both sides, we get } \sin^2 \theta = 9(1 - \sin^2 \theta)$$

$$\Rightarrow 10 \sin^2 \theta = 9 \Rightarrow \sin \theta = \sqrt{\frac{9}{10}} = -\frac{3}{\sqrt{10}}$$

($\because \theta$ lies in IIIrd quadrant, so $\sin \theta$ will be negative)

41. (a) $\sin 20^\circ \cos 70^\circ + \cos 20^\circ \sin 70^\circ$

$$= \sin(20^\circ + 70^\circ) = \sin 90^\circ = 1$$

$$[\because \sin A \cos B + \cos A \sin B = \sin(A + B)]$$

42. (a) $\phi + \theta = 180^\circ \Rightarrow \theta = 180^\circ - \phi \Rightarrow \sin \theta = \sin(180^\circ - \phi)$

$$\sin \theta = \sin \phi$$

43. (b) $\sqrt{\cos x \operatorname{cosec} y - \cos x \sin y}$

$$= \sqrt{\cos x \cdot \operatorname{cosec}(90^\circ - x) - \cos x \cdot \sin(90^\circ - x)}$$

($\because x + y = 90^\circ$, given)

$$= \sqrt{\cos x \cdot \sec x - \cos^2 x}$$

$$= \sqrt{1 - \cos^2 x} = \sqrt{\sin^2 x} = \sin x$$

44. (c) $\sqrt{2 + \sqrt{2 + 2 \cos 4\theta}} = \sqrt{2 + \sqrt{2(1 + \cos 4\theta)}}$

$$= \sqrt{2 + \sqrt{2(2 \cos^2 2\theta)}} = \sqrt{2 + \sqrt{4 \cos^2 2\theta}} = \sqrt{2 + 2 \cos 2\theta}$$

$$= \sqrt{2(1 + \cos 2\theta)} = \sqrt{2(2 \cos^2 \theta)} = \sqrt{4 \cos^2 \theta} = 2 \cos \theta$$

45. (b) Given, $\tan A = \frac{5}{6}$ and $\tan B = \frac{1}{11}$

$$\therefore \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\frac{5}{6} + \frac{1}{11}}{1 - \frac{5}{6} \cdot \frac{1}{11}} = \frac{\frac{55 + 6}{66}}{\frac{66 - 5}{66}} = \frac{61}{61} = 1$$

$$\Rightarrow \tan(A + B) = 1 \Rightarrow \tan(A + B) = \tan \frac{\pi}{4} \Rightarrow A + B = \frac{\pi}{4}$$

46. (c) $\frac{1}{\sqrt{2}} (\cos A - \sin A) = \frac{\cos A}{\sqrt{2}} - \frac{\sin A}{\sqrt{2}}$

$$= \cos \frac{\pi}{4} \cos A - \sin \frac{\pi}{4} \sin A = \cos \left(\frac{\pi}{4} + A \right)$$

$$[\because \cos A \cos B - \sin A \sin B = \cos(A + B)]$$

47. (d) $\sin \frac{\pi}{4} \cdot \cos \frac{\pi}{12} - \cos \frac{\pi}{4} \cdot \sin \frac{\pi}{12} = \sin \left(\frac{\pi}{4} - \frac{\pi}{12} \right)$

$$[\because \sin A \cos B - \cos A \sin B = \sin(A - B)]$$

$$= \sin \left(\frac{\pi}{6} \right) = \frac{1}{2}$$

48. (c) Given, $\tan \theta = \frac{x \sin \phi}{1 - x \cos \phi} \Rightarrow x \sin \phi = \tan \theta - x \cos \phi \tan \theta$

$$\Rightarrow x = \frac{\tan \theta}{\sin \phi + \cos \phi \tan \theta} = \frac{\sin \theta}{\cos \theta \sin \phi + \cos \phi \sin \theta} = \frac{\sin \theta}{\sin(\theta + \phi)}$$

Similarly, $y = \frac{\sin \phi}{\sin(\theta + \phi)}$, Hence, $\frac{x}{y} = \frac{\sin \theta}{\sin \phi}$

49. (b) Given, $\sin(\theta + \phi) = 2 \sin(\theta - \phi)$

$$\Rightarrow \sin \theta \cos \phi + \cos \theta \sin \phi = 2(\sin \theta \cos \phi - \cos \theta \sin \phi)$$

$$\Rightarrow 3 \cos \theta \sin \phi = \sin \theta \cos \phi \Rightarrow \frac{3 \sin \phi}{\cos \phi} = \frac{\sin \theta}{\cos \theta}$$

$$\Rightarrow 3 \tan \phi = \tan \theta \text{ or } \tan \theta = 3 \tan \phi$$

50. (a) Given, $\sin(\theta + \alpha) = \cos(\theta + \alpha) \Rightarrow \frac{\sin(\theta + \alpha)}{\cos(\theta + \alpha)} = 1$

$$\Rightarrow \tan(\theta + \alpha) = 1 \Rightarrow \frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} = 1$$

$$\Rightarrow \tan \theta + \tan \alpha = 1 - \tan \theta \tan \alpha$$

$$\Rightarrow \tan \theta(1 + \tan \alpha) = 1 - \tan \alpha \Rightarrow \tan \theta = \frac{1 - \tan \alpha}{1 + \tan \alpha}$$

51. (c) $\cos^2 \frac{\theta - \phi}{2} - \sin^2 \frac{\theta + \phi}{2}$

$$= \cos \left(\frac{\theta - \phi}{2} + \frac{\theta + \phi}{2} \right) \cdot \cos \left(\frac{\theta - \phi}{2} - \frac{\theta + \phi}{2} \right)$$

$$[\because \cos^2 A - \sin^2 B = \cos(A + B) \cos(A - B)]$$

$$= \cos \theta \cdot \cos(-\phi) = \cos \theta \cdot \cos \phi$$

52. (c) $x^2 + y^2 + z^2 = r^2 \sin^2 \theta \cos^2 \phi + r^2 \sin^2 \theta \sin^2 \phi + r^2 \cos^2 \theta$

$$= r^2 \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + r^2 \cos^2 \theta$$

$$= r^2 \sin^2 \theta + r^2 \cos^2 \theta = r^2 (\sin^2 \theta + \cos^2 \theta) = r^2$$

53. (c) Given, $\cot \theta = \frac{7}{24}$, $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta = 1 + \frac{49}{576} = \frac{625}{576}$

$$\therefore \operatorname{cosec} \theta = \pm \frac{25}{24}$$

$$\Rightarrow \sin \theta = \pm \frac{24}{25}, \cos^2 \theta = 1 - \sin^2 \theta = 1 - \frac{576}{625} = \frac{49}{625}$$

$$\therefore \cos \theta = \pm \frac{7}{25}, \text{ As } \pi < \theta < \frac{3\pi}{2}$$

$$\therefore \sin \theta \text{ and } \cos \theta \text{ both are negative.}$$

$$\therefore \sin \theta = \frac{-24}{25}, \cos \theta = \frac{-7}{25}$$

$$\therefore \cos \theta - \sin \theta = \frac{-7}{25} + \frac{24}{25} = \frac{17}{25}$$

54. (b) Given, $\sin \alpha = \frac{1}{2} = \sin 30^\circ \Rightarrow \alpha = 30^\circ$

Also, $\beta = \frac{\pi}{2} - \alpha = 90^\circ - 30^\circ = 60^\circ$

55. (c) Given, $\sin x - \cos x = 0 \Rightarrow \sin x = \cos x \Rightarrow \tan x = 1$

$$\Rightarrow \tan x = \tan \frac{\pi}{4} \Rightarrow x = \frac{\pi}{4}$$

$$\therefore \sin^4 x + \cos^4 x = \sin^4 \frac{\pi}{4} + \cos^4 \frac{\pi}{4} = \left(\frac{1}{\sqrt{2}} \right)^4 + \left(\frac{1}{\sqrt{2}} \right)^4$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

56. (b) Given, $3 \cos x = 5 \sin x \Rightarrow \frac{\sin x}{\cos x} = \frac{3}{5} \Rightarrow \tan x = \frac{3}{5}$

$$\frac{5 \sin x - 2 \sec^3 x + 2 \cos x}{5 \sin x + 2 \sec^3 x - 2 \cos x} = \frac{5 \tan x - 2 \sec^4 x + 2}{5 \tan x + 2 \sec^4 x - 2}$$

(dividing numerator and denominator by $\cos x$)

$$= \frac{5 \tan x - 2(1 + \tan^2 x)^2 + 2}{5 \tan x + 2(1 + \tan^2 x)^2 - 2}$$

$$= \frac{5 \times \frac{3}{5} - 2 \left(1 + \frac{9}{25} \right)^2 + 2}{5 \times \frac{3}{5} + 2 \left(1 + \frac{9}{25} \right)^2 - 2} = \frac{3 + 2 - 2 \times \frac{1156}{625}}{3 - 2 + 2 \times \frac{1156}{625}}$$

$$= \frac{3125 - 2312}{625 + 2312} = \frac{813}{2937} = \frac{271}{979}$$

57. (b) $\sqrt{\frac{1 + \cos x}{1 - \cos x}} = \sqrt{\frac{(1 + \cos x)}{(1 - \cos x)} \times \frac{1 + \cos x}{1 + \cos x}}$

$$= \frac{1 + \cos x}{\sin x} = \frac{1}{\sin x} + \frac{\cos x}{\sin x} = \operatorname{cosec} x + \cot x$$

58. (a) $\cos^4 x - \sin^4 x = (\cos^2 x - \sin^2 x)(\sin^2 x + \cos^2 x)$

$$= [(\cos^2 x) - (1 - \cos^2 x)] = 2 \cos^2 x - 1 \quad (\because \sin^2 x + \cos^2 x = 1)$$

59. (a) $\frac{\sin \theta}{1 + \cos \theta} = \frac{\sin \theta}{1 + \cos \theta} \times \frac{1 - \cos \theta}{1 - \cos \theta} = \frac{\sin \theta(1 - \cos \theta)}{\sin^2 \theta} = \frac{1 - \cos \theta}{\sin \theta}$

60. (a) $\frac{5 \cos \theta - 4}{3 - 5 \sin \theta} - \frac{3 + 5 \sin \theta}{4 + 5 \cos \theta}$

$$= \frac{(5 \cos \theta - 4)(4 + 5 \cos \theta) - (3 + 5 \sin \theta)(3 - 5 \sin \theta)}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)}$$

$$= \frac{(25 \cos^2 \theta - 16) - (9 - 25 \sin^2 \theta)}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)}$$

$$= \frac{25(\sin^2 \theta + \cos^2 \theta) - 25}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)} = \frac{25 - 25}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)} = 0$$

61. (a) $\sin(\alpha + \beta) = \sqrt{1 - \cos^2(\alpha + \beta)} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$

$$\cos(\alpha - \beta) = \sqrt{1 - \sin^2(\alpha - \beta)} = \sqrt{1 - \frac{25}{169}} = \sqrt{\frac{144}{169}} = \frac{12}{13}$$

$$\therefore \tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \left(\frac{3}{5} \times \frac{5}{4} \right) = \frac{3}{4}$$

$$\tan(\alpha - \beta) = \frac{\sin(\alpha - \beta)}{\cos(\alpha - \beta)} = \frac{5}{13} \times \frac{13}{12} = \frac{5}{12}$$

$$\therefore \tan(2\alpha) = \tan[(\alpha + \beta) + (\alpha - \beta)]$$

$$= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta) \cdot \tan(\alpha - \beta)} = \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \times \frac{5}{12}} = \frac{56}{33}$$

62. (a) $\tan(45^\circ + x) = \frac{\tan 45^\circ + \tan x}{1 - \tan 45^\circ \tan x} = \frac{1 + \tan x}{1 - \tan x}$

$$\tan(45^\circ - x) = \frac{\tan 45^\circ - \tan x}{1 + \tan 45^\circ \tan x} = \frac{1 - \tan x}{1 + \tan x}$$

$$\therefore \frac{\tan(45^\circ + x)}{\tan(45^\circ - x)} = \frac{\frac{1 + \tan x}{1 - \tan x}}{\frac{1 - \tan x}{1 + \tan x}} = \left[\frac{1 + \tan x}{1 - \tan x} \right]^2$$

$$\begin{aligned}
 63. (a) & \frac{(1 - \sin A \cos A)(\sin^2 A - \cos^2 A)}{\cos A (\sec A - \operatorname{cosec} A)(\sin^3 A + \cos^3 A)} \\
 &= \frac{(1 - \sin A \cos A)(\sin^2 A - \cos^2 A)}{\left[\cos A \left(\frac{1}{\cos A} - \frac{1}{\sin A} \right) (\sin A + \cos A) \right]} \\
 &= \frac{[(1 - \sin A \cos A)(\sin A - \cos A)]}{[\cos A (\sin A - \cos A)(\sin A + \cos A)]} = \sin A
 \end{aligned}$$

$$\begin{aligned}
 64. (c) & (1 + \tan A)(1 + \tan B) \\
 &= (1 + \tan A)[1 + \tan(45^\circ - A)] \\
 &= (1 + \tan A) \left[1 + \frac{\tan 45^\circ - \tan A}{1 + \tan 45^\circ \tan A} \right] \quad (\because A + B = 45^\circ) \\
 &= (1 + \tan A) \left[1 + \frac{1 - \tan A}{1 + \tan A} \right] \\
 &= (1 + \tan A) \left[\frac{1 + \tan A + 1 - \tan A}{1 + \tan A} \right] \\
 &= (1 + \tan A) \left[\frac{2}{1 + \tan A} \right] = 2
 \end{aligned}$$

$$\begin{aligned}
 65. (a) & \because \cos A - \cos B = 2 \sin \frac{(A+B)}{2} \cdot \sin \frac{(B-A)}{2} \\
 \text{and } \sin A + \sin B &= 2 \sin \frac{(A+B)}{2} \cdot \cos \frac{(A-B)}{2} \\
 \therefore \frac{\cos 2B - \cos 2A}{\sin 2B + \sin 2A} &= \frac{2 \sin(A+B) \sin(A-B)}{2 \sin(A+B) \cos(A-B)} = \tan(A-B)
 \end{aligned}$$

$$\begin{aligned}
 66. (d) & \sec^2 x + \tan^2 x = 7 \quad (\because \sec^2 x = 1 + \tan^2 x) \\
 1 + \tan^2 x + \tan^2 x &= 7 \Rightarrow 1 + 2 \tan^2 x = 7 \\
 \Rightarrow 2 \tan^2 x &= 6 \Rightarrow \tan x = \sqrt{3} = \tan 60^\circ \\
 \therefore x &= 60^\circ
 \end{aligned}$$

$$\begin{aligned}
 67. (a) & \cos 3\theta + \sin 3\theta = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos 3\theta + \frac{1}{\sqrt{2}} \sin 3\theta \right) \\
 &= \sqrt{2} (\sin 45^\circ \cos 3\theta + \cos 45^\circ \sin 3\theta) = \sqrt{2} [\sin(45^\circ + 3\theta)] \\
 \cos 3\theta + \sin 3\theta &\text{ is maximum when } \sin(45^\circ + 3\theta) \text{ is maximum} \\
 \text{The maximum value of } \sin(45^\circ + 3\theta) &= 1 = \sin 90^\circ \\
 \Rightarrow 45^\circ + 3\theta &= 90^\circ \Rightarrow \theta = 15^\circ
 \end{aligned}$$

$$\begin{aligned}
 68. (c) & \sin^2 x + \sin^2 y + \sin^2 z = (\sin x + \sin y + \sin z)^2 \\
 \Rightarrow \sin x \sin y + \sin y \sin z + \sin z \sin x &= 0 \\
 [\because (a+b+c)^2 &= a^2 + b^2 + c^2 + 2(ab+bc+ca)] \\
 \text{On dividing both sides by } \sin x \sin y \sin z & \\
 \Rightarrow \frac{1}{\sin x} + \frac{1}{\sin y} + \frac{1}{\sin z} &= 0
 \end{aligned}$$

$$\begin{aligned}
 69. (d) & \text{ Given, } \sin x + \sin^2 x = 1 \quad \dots(i) \\
 \Rightarrow \sin x &= 1 - \sin^2 x = \cos^2 x \quad (\because \sin^2 x + \cos^2 x = 1) \\
 \therefore \cos^2 x + \cos^4 x &= (\sin x) + \sin^2 x = 1 \quad [\text{from Eq. (i)}]
 \end{aligned}$$

$$\begin{aligned}
 70. (a) & (\cos 35^\circ + \cos 85^\circ) + \cos 155^\circ \\
 &= 2 \cos 60^\circ \cos 25^\circ + \cos(180^\circ - 25^\circ) \\
 &= 2 \times \frac{1}{2} \cos 25^\circ - \cos 25^\circ = \cos 25^\circ - \cos 25^\circ = 0
 \end{aligned}$$

71. (a) Refer to example 16.

$$\begin{aligned}
 72. (d) & \text{ Given, } \tan \theta = \frac{p}{q} \\
 p \cos 2\theta + q \sin 2\theta &= p \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right) + q \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) \\
 &= p \left(\frac{1 - \frac{p^2}{q^2}}{1 + \frac{p^2}{q^2}} \right) + q \left(\frac{\frac{2p}{q}}{1 + \frac{p^2}{q^2}} \right) = p \left(\frac{q^2 - p^2}{q^2 + p^2} \right) + \left(\frac{2pq^2}{q^2 + p^2} \right) \\
 &= \frac{p(q^2 - p^2 + 2q^2)}{p^2 + q^2} = \frac{p(3q^2 - p^2)}{p^2 + q^2}
 \end{aligned}$$

$$73. (c) \text{ Given, } x = y \cos \frac{2\pi}{3} = z \cos \frac{4\pi}{3} \Rightarrow x = \frac{-y}{2} = \frac{-z}{2} = k \quad (\text{let})$$

$$\begin{aligned}
 x &= k, y = -2k, z = -2k \\
 \Rightarrow xy + yz + zx &= k(-2k) + (-2k)(-2k) + (-2k)k \\
 &= -2k^2 + 4k^2 - 2k^2 = 0
 \end{aligned}$$

$$\begin{aligned}
 74. (c) & \text{ Given, } \sin \theta + \cos \theta = m \quad \dots(i) \\
 \text{and } \sec \theta + \operatorname{cosec} \theta &= n \Rightarrow \frac{1}{\cos \theta} + \frac{1}{\sin \theta} = n
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \frac{\sin \theta + \cos \theta}{\sin \theta \cos \theta} &= n \\
 \Rightarrow \frac{m}{\sin \theta \cos \theta} &= n \Rightarrow \sin \theta \cos \theta = \frac{m}{n} \quad [\text{from Eq. (i)}]
 \end{aligned}$$

$$\begin{aligned}
 \text{On squaring Eq. (i), we get } \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta &= m^2 \\
 1 + 2 \frac{m}{n} &= m^2 \Rightarrow \frac{2m}{n} = m^2 - 1 \quad [\because \sin^2 \theta + \cos^2 \theta = 1] \\
 \Rightarrow 2m &= (m^2 - 1)n
 \end{aligned}$$

$$\begin{aligned}
 75. (b) & \text{ Given, } \operatorname{cosec} \theta - \sin \theta = a^3 \Rightarrow \frac{1}{\sin \theta} - \sin \theta = a^3 \\
 \Rightarrow \frac{1 - \sin^2 \theta}{\sin \theta} &= a^3 \Rightarrow \frac{\cos^2 \theta}{\sin \theta} = a^3 \Rightarrow \cos^2 \theta = a^3 \sin \theta \quad \dots(ii)
 \end{aligned}$$

$$\begin{aligned}
 \text{and } \sec \theta - \cos \theta &= b^3 \Rightarrow \frac{1}{\cos \theta} - \cos \theta = b^3 \Rightarrow \frac{1 - \cos^2 \theta}{\cos \theta} = b^3 \\
 \Rightarrow \frac{\sin^2 \theta}{\cos \theta} &= b^3 \Rightarrow \sin^2 \theta = b^3 \cos \theta \quad \dots(iii)
 \end{aligned}$$

On putting $\cos \theta = \frac{\sin^2 \theta}{b^3}$ in Eq. (i), we get

$$\begin{aligned}
 \frac{\sin^4 \theta}{b^6} &= a^3 \sin \theta \Rightarrow \sin^3 \theta = a^3 b^6 \\
 \therefore \sin \theta &= ab^2 \quad \dots(iv) \\
 \text{Similarly, } \cos \theta &= a^2 b
 \end{aligned}$$

On squaring and adding Eqs. (iii) and (iv), we get

$$1 = a^2 b^4 + a^4 b^2 \Rightarrow 1 = a^2 b^2 (a^2 + b^2)$$

76. (d) As, we know that $\cos \theta \geq -1$ and $\cos \theta \leq 1$

$$\begin{aligned}
 \therefore \cos 2x &\geq -1 \Rightarrow 1 + \cos 2x \geq 0 \\
 \cos 2x &\leq 1 \Rightarrow 1 + \cos 2x \leq 2
 \end{aligned}$$

So, minimum value of $1 + \cos 2x$ is 0 and maximum value of $1 + \cos 2x$ is 2.